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Project Report

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Circuit description:

The automatic fire extinguisher system consists of the following key components:

- **IR receiver:** Detects infrared radiation from fire and sends a signal to the transistor.
- **BC547 transistor:** Acts as a switch. When triggered by the IR receiver, it turns on, allowing current to flow.
- **Resistors:**
 - 220 Ω resistor protects the red led.
 - 1 k Ω resistor stabilizes the transistor base.
 - 10 k Ω resistor sets correct transistor biasing.
- **Green LED:** Lights up when the system detects fire, indicating that the pump is activated.
- **Water pump:** Turns on when the transistor is activated, spraying water to extinguish the fire.
- **Power source (3.7v battery):** Provides power to the entire system.

Working:

No fire: The system stays inactive.

Fire detected: The IR receiver triggers the transistor, activating the water pump and lighting the green led.

Real time diagram-1:

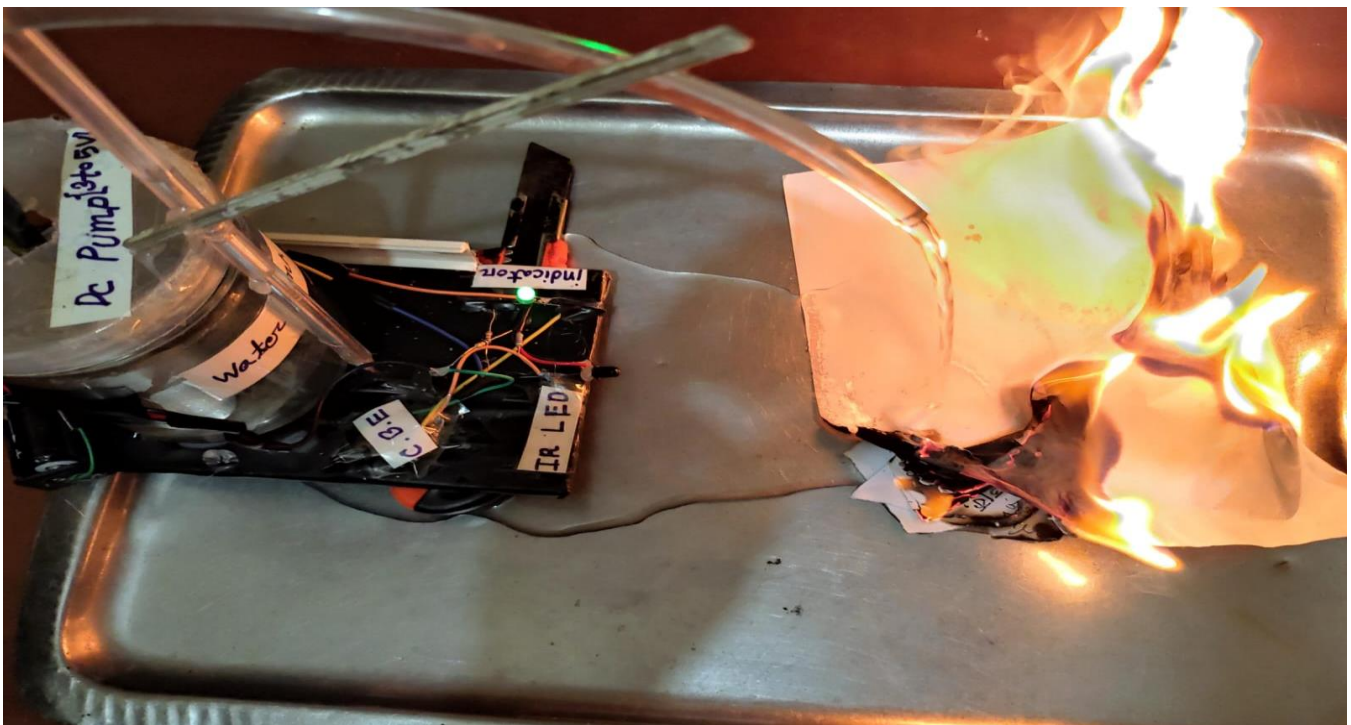


Figure 2: Detected fire and started spreading water.

Description:

The Automatic Fire Extinguisher System continuously monitors for fire using an IR receiver. When flames are detected, the IR receiver sends a signal to activate the BC547 transistor, which powers the water pump. The pump sprays water to extinguish the fire, and the green LED lights up as an indicator.

Real time diagram-2:

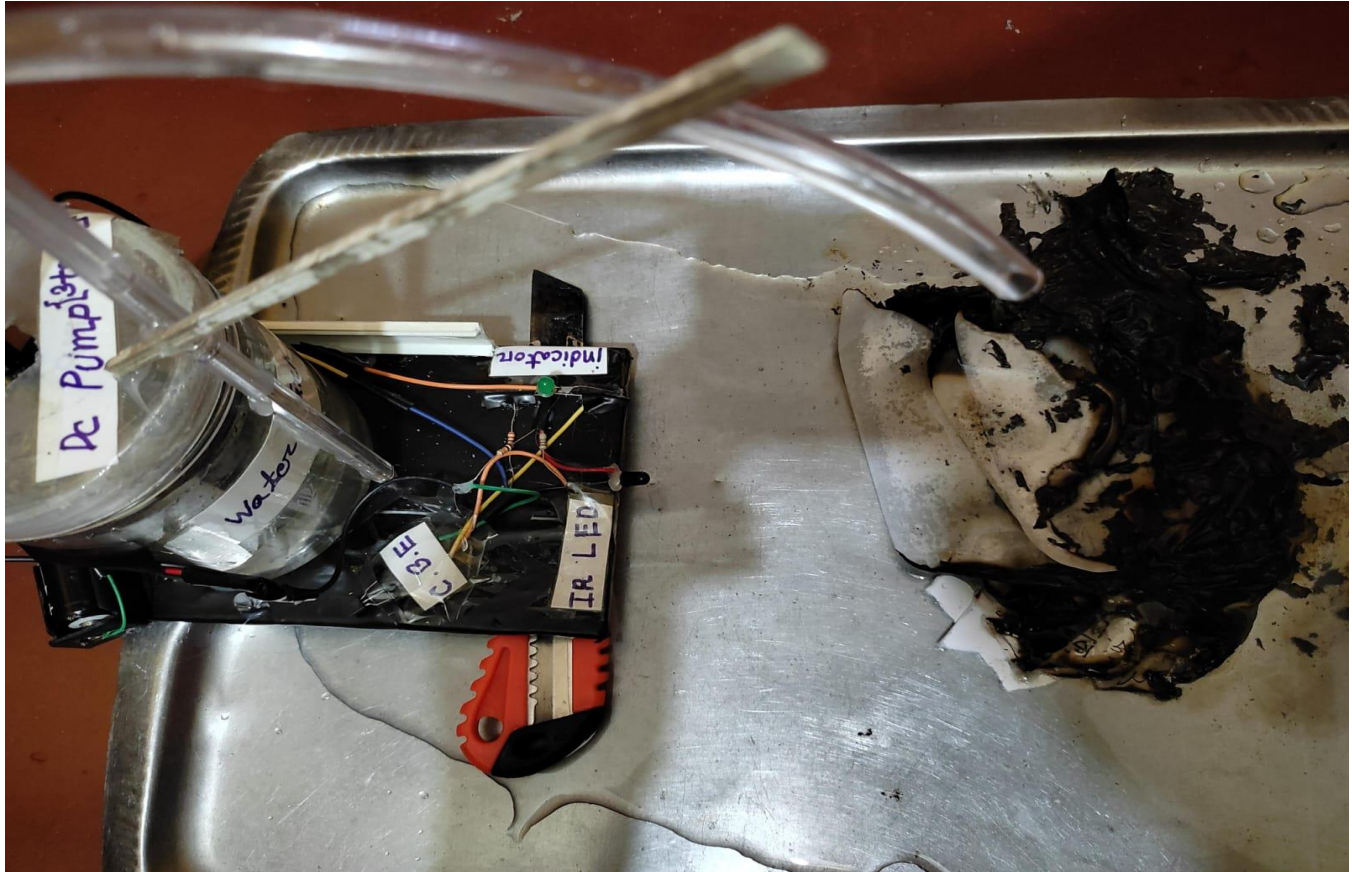


Figure 3: After extinguishing the fire.

Description:

Once the fire is fully extinguished, the system seamlessly transitions back to its idle state. This automated process eliminates the need for manual intervention, allowing the system to remain on standby and ready for the next potential incident. By efficiently shutting down after its task is completed, the system conserves energy and ensures smooth, hands-free operation, providing both reliability and convenience.

Working principle:

The automatic fire extinguisher system works by using an IR sensor to detect fire. When no fire is present, the system stays off. If fire is detected, the sensor sends a signal to a transistor, which turns on the water pump and lights up a green LED to show the system is working. The water pump sprays water to put out the fire. Once the fire is extinguished, the system automatically goes back to standby mode, ready to respond again. This makes it a quick and simple solution for small fire emergencies without needing human help.

Costing:

Resistors 9 TK (3-piece), Motor 50 TK, Green led 2 TK, IR receiver led 5 TK, Battery 50 TK, Wire 6 TK
Total = 122 TK

Discussion:

The conclusion of this automatic fire extinguisher system is an innovative and cost-effective solution for fire safety, designed to automatically detect and respond to fires by activating a water pump to extinguish flames. The project leverages simple electronic components like an IR receiver, BC547 transistor, and water pump, making it an accessible and functional prototype for basic fire suppression.

Advantages:

Automatic operation: Eliminates the need for human intervention, allowing the system to respond to fires quickly.

Cost-effective: Utilizes low-cost, easily available components, making it affordable for small-scale applications.

Quick response: Helps in preventing the spread of fire by immediately triggering water discharge.

Simplicity: The circuit is straightforward, easy to understand, and implement, making it ideal for educational purposes and basic fire safety demonstrations.

Visual indicator: The red led provides clear feedback that the system is active and responding to fire.

Disadvantages:

Limited scope: The system works in a small-scale environment and may not be effective for large or high-intensity fires.

Water use: The water pump can only provide limited water and is not suited for extensive firefighting.

Dependence on IR receiver: The IR sensor may not detect all types of fire or may be affected by environmental conditions such as smoke or light interference.

Power source: The 3.7v battery may not provide enough power for extended periods, requiring regular maintenance or charging.

Detections: The system only detects fire via infrared radiation and does not account for smoke, which could also be an indicator of fire.

Mapping:

K3: Theoretical knowledge from **Study of Common Emitter (CE) Amplifier**.

A Common Emitter (CE) amplifier is a transistor-based circuit where the emitter terminal is common to both input and output. It amplifies current, with a small base current variation producing a larger collector current. The output signal is inverted (180° phase shift) and exhibits high voltage and current gain, moderate input resistance, and high output resistance. CE amplifiers are widely used in signal amplification, oscillators, and analog processing. Advantages include high gain, while limitations include phase inversion and distortion at high amplitudes. Key experimental goals include measuring voltage and current gain, input/output resistance, and frequency response.

K5: Project Description (**Automatic Fire Extinguisher System**):

This project creates an automated system to detect and extinguish fires using an IR sensor and a water pump. When flames are detected, the sensor activates the pump to spray water and a green LED light up to indicate operation. The system is simple, cost-effective, and ideal for small-scale applications, offering quick, hands-free fire suppression. However, it has limitations in large-scale or high-intensity fires due to its reliance on basic components and limited water capacity.

K6: The **Automatic Fire Extinguisher System** uses the following technologies:

- **Infrared (IR) Sensor Technology:** Detects flames by sensing infrared radiation emitted by fire.
- **Electronics and Switching Technology:** Utilizes a BC547 transistor to act as a switch, controlling the water pump.
- **Micro-Scale Actuation:** Employs a small water pump (3V–5V) to spray water when activated.
- **Battery-Powered Circuitry:** Operates using a 3.7V battery, enabling portability and independence from external power sources.
- **Indicator System:** Includes an LED for visual feedback when the system is active.

A2: Based on the type of problem the project is created:

The project is created to address the problem of fire hazards in small-scale environments like homes, offices, or workshops. Fires can lead to significant loss of life and property if not addressed quickly. This system offers a low-cost, automated solution to detect and extinguish fires promptly, minimizing response time and eliminating the need for human intervention during emergencies.

The problem being tackled is:

- **Delay in fire response:** Manual firefighting may take time and put lives at risk.
- **Lack of affordable fire safety solutions:** Existing systems may be expensive or unsuitable for small-scale use.